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Corresponding author:
L. S. Wright;
Email: luka.wright@research.uwa.edu.au

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Plants in limbo: The theory of detrital photosynthesis

Luka Seamus Wright[1,2]

[1]Oceans Institute, University of Western Australia, Perth, Australia; [2]School of Biological Sciences, University of Western Australia, Perth, Australia

Abstract

Kelp detritus can maintain photosynthesis for months following detachment. This may be explained by limited tissue differentiation in non-vascular plants, but it remains unclear if detrital photosynthesis is restricted to macroalgae. Here I formalise the theory of detrital photosynthesis by developing a mathematical model and performing a meta-analysis of post-detachment physiology across 129 studies on 92 species of terrestrial and aquatic plants, including original data on the seagrasses *Amphibolis antarctica* and *Halophila ovalis*. Detrital photosynthesis is a general phenomenon, but its longevity in macroalgae is unique. Median photosynthetic half-life increases from terrestrial plants (11 days) over freshwater Alismatales (14 days) and seagrasses (17 days) to seaweeds (44 days) and only matches or exceeds detrital half-life for seaweeds (3:1). Detrital photosynthesis therefore determines macroalgal but not vascular plant decomposition. The resulting causal model explains why light slows kelp decomposition, contrary to the expected effect of photodegradation. The presented theory questions conventional understanding of decomposition and the carbon cycle.

1. Introduction

> . . . tossed and swept like dead leaves from one spot to another, never resting, never giving back their goodness to the earth, never fully dead but in some vast limbo between life and extinction, tossing and tumbling without end . . . – Philip Pullman (1994) The Tin Princess.

Seaweed detrital dynamics are influenced by detrital photosynthesis. Detrital photosynthesis is defined as the persistence of photosynthesis after detachment (Frontier et al., 2021; Wright et al., 2022, 2024). Kelp detritus can remain photosynthetic for months (de Bettignies et al., 2020; Frontier et al., 2021, 2022; Wright et al., 2022, 2024; Wright & Foggo, 2021; Wright & Kregting, 2023) and seaweed rafts sink before they stop photosynthesising (Graiff et al., 2013, 2016; Rothäusler et al., 2020; Tala et al., 2019). This results in atypical decomposition trajectories with a distinct lag in the onset of decay (Birch et al., 1983; Bourgués et al., 1996; Brouwer, 1996; Frontier et al., 2021, 2022; Kennedy & Blain, 2025; Vandendriessche et al., 2007). The fundamental effect on detrital dynamics becomes apparent when comparing live with dead or senescent seaweed detritus, which decomposes up to an order of magnitude faster (Birch et al., 1983; Brouwer, 1996; de Bettignies et al., 2020; Hamersley et al., 2015; White & Norkko, 2025). This evidence stands in stark contrast to the established ecological definition of detritus as 'dead organic matter' (Moore et al., 2004, discussed in Wright & Kregting, 2023). One solution to this conundrum is to insist that detached plant material is not detritus as long as it is still alive. However, detached plants (except for holopelagic and pleustophytic species) are commonly referred to as detritus. For most plants, detachment provides a clear demarcation of the detrital phase, which the ensuing photosynthetic limbo does not. By way of compromise, I continue to use the term 'detritus' for any detached plant tissue and 'detritus sensu stricto' for detached *and* dead plant tissue.

Detrital photosynthesis underpins other aspects of physiology. Remarkably, seaweed may continue to grow after detachment (Filbee-Dexter et al., 2022; Frontier et al., 2021, 2022; Norton & Mathieson, 1983; Pedersen et al., 2021; Rothäusler et al., 2011, 2018; Tala et al., 2016; Vandendriessche et al., 2007; Wright et al., 2022). This suggests that detrital photosynthesis can counteract decomposition by supporting tissue growth and repair. Moreover, chemical defences are maintained after detachment. Excised seaweed tissues are capable of oxidative bursts (Küpper et al., 2001, 2002; Thomas et al., 2014; Weinberger et al., 1999; Weinberger & Friedlander, 2000) and constitutive hydrogen peroxide release (Küpper et al., 2002; Twigg et al., 2020), which inhibit heterotrophic bacteria (Küpper et al., 2001, 2002; Twigg et al., 2020; Weinberger & Friedlander, 2000) and grazers (Thomas et al., 2014). Excised brown algal tissues additionally produce phlorotannins, which aid wound healing besides acting as chemical defence (Halm et al., 2011; Koivikko et al., 2005; Lüder & Clayton, 2004). Steady and increasing phlorotannin content of kelp

detritus (Tala et al., 2013, 2016, Tala et al., 2019, de Bettignies et al., 2020, Wright et al., 2022, but see Pedersen et al., 2021) may thus be interpreted as a response to decomposition (Wright et al., 2022). This suggests that detrital photosynthesis can also counteract decomposition by enabling chemical defence. Finally, there is overwhelming evidence for reproductive kelp detritus (Fraser et al., 2018; Hernández-Carmona et al., 2006; Macaya et al., 2005; Rothäusler et al., 2011; Tala et al., 2013) and persistence or development of reproductive structures in experimentally detached seaweeds (de Bettignies et al., 2020; Frontier et al., 2021; Hernández-Carmona et al., 2006; Rothäusler et al., 2011, 2018, 2020; Tala et al., 2016, 2019). Detritus can therefore act as a propagule, often better suited to long-distance dispersal than spores and gametes (Fraser et al., 2018; Norton, 1992). Even seaweed detritus that is not reproductive can be a means of dispersal through vegetative propagation (Norton & Mathieson, 1983), which is also common among seagrasses (Hall et al., 2006; Hanuise et al., 2026; Stafford-Bell et al., 2015; Weatherall et al., 2025). Detrital photosynthesis, providing the energy allocated to growth, repair, defence and reproduction, must thus be regarded as evidence of detrital physiology more broadly.

There is currently no reference point for detrital photosynthesis. The term was first introduced quite recently in the context of kelp decomposition (Frontier et al., 2021; Wright et al., 2022, 2024). But there was clearly awareness of the phenomenon among early phycologists who noted the existence of 'living litter' (Norton & Mathieson, 1983) and ability of excised tissue to maintain photosynthesis (Drew, 1983), and deemed it necessary to 'pre-kill' detritus to measure decomposition (Birch et al., 1983; Brouwer, 1996; Josselyn & Mathieson, 1980; Rice & Tenore, 1981; Smith & Foreman, 1984). The term 'detrital photosynthesis' is not used in other fields of botany but the amount of research into photosynthesis of excised leaves of terrestrial and freshwater plants is enormous (e.g. Gauthier & Jacobs, 2018; Jana & Choudhuri, 1980, 1982; Kar et al., 2021; Kar & Choudhuri, 1986, 1987; Powles et al., 2006; Thimann & Satler, 1979). The literature also includes three studies on seagrasses (Knauer & Ayers, 1977; Pellikaan, 1982; Strother & Vatta, 1986), the only fully marine angiosperms and vascular counterparts to seaweeds (Olsen et al., 2016). The aims of measuring photosynthesis in detached leaves are varied, with most studies using excision as an experimental proxy for plant senescence. No attempt has been made to align these seemingly incongruous data in a formal comparison across plant groups (sensu Bolton, 2016). It consequently remains unclear if detrital photosynthesis is unique to seaweeds. Although it has been suggested that detrital photosynthesis may be explained by the ability of all macroalgal tissues to independently photosynthesise to some extent (Wright & Kregting, 2023), there is no certainty due to lack of comparison with other plants. The mechanism underlying the ability of seaweeds to maintain detrital photosynthesis remains unclear. It could simply be their aquatic nature, which removes reliance on roots for water and nutrient provisioning. Or it could be more deeply rooted in their non-vascularity, which requires minimal tissue differentiation. To understand the prerequisites for detrital photosynthesis and thus predict prevalence among plants, we need to look beyond seaweeds.

Seagrasses are suitable candidates for further investigation of detrital photosynthesis. While they are also marine macrophytes, seagrasses are fundamentally different from seaweeds. Seagrasses are vascular plants which, due to their terrestrial ancestry (Olsen et al., 2016), are differentiated into roots, rhizomes, stems and leaves. However, convergent evolution with seaweeds (Olsen et al., 2016) has reduced their reliance on roots for nutrient uptake. Seagrass leaves tend to outperform their roots in supplying nutrients to the plant (Gras et al., 2003; Pedersen et al., 1997; Pedersen & Borum, 1992; Stapel et al., 1996; Viana et al., 2019). This suggests that seagrass leaves can independently photosynthesise, seemingly reducing the primary function of the roots to anchorage, analogous to the holdfast of seaweeds. However, despite the leaves' greater uptake affinity, roots can supply more nitrogen because of the higher ammonium concentration in sediment pore water (Short & McRoy, 1984, Lee & Dunton, 1999, but see Pedersen & Borum, 1992) and tap into nutrient pools that are unavailable to leaves, such as particulate (Evrard et al., 2005) and microbially fixed (Mohr et al., 2021; Patriquin & Knowles, 1972) nitrogen. Nutrients are also translocated from roots to leaves (Short & McRoy, 1984; Viana et al., 2019), even those tens of centimetres from the source (Marbà et al., 2002). Clearly the photosynthetic independence of seagrass leaves is at a halfway point between terrestrial plants and macroalgae, so seagrasses probably display detrital photosynthesis to some extent. Relatively slow and light-dependent chlorophyll degradation in seagrass detritus (Knauer & Ayers, 1977; Pellikaan, 1982; Strother & Vatta, 1986) provides some support for this hypothesis. Seagrass detritus also often incudes green leaves and vegetative fragments (Jiménez-Ramos et al., 2023; Knauer & Ayers, 1977; Ochieng & Erftemeijer, 1999; Stafford-Bell et al., 2015; Weatherall et al., 2025), likely owing to premature detachment (Cebrián et al., 1997; Hall et al., 2006; Jiménez-Ramos et al., 2023; Weatherall et al., 2025) and low nutrient resorption prior to abscission (Harrison, 1989; Hemminga et al., 1999; Pedersen et al., 1997; Pedersen & Borum, 1992; Rosch & Koch, 2009). The fragments are typically viable and may persist for a few weeks (Hall et al., 2006; Hanuise et al., 2026; Stafford-Bell et al., 2015; Weatherall et al., 2025). Nonetheless, seagrass detrital photosynthesis remains to be directly investigated.

Here I aim to formalise the theory of detrital photosynthesis. I hope that this will provide a better conceptual foundation for the investigation of plant decomposition and carbon cycling more generally. To address this aim, I approached detrital photosynthesis holistically for the first time. I conducted a brief ex situ decomposition experiment with seagrass leaves to fill the knowledge gap regarding seagrass detrital photosynthesis. As diverse representatives of seagrasses, I chose the dugong grass *Halophila ovalis* (Hydrocharitaceae) and sea nymph *Amphibolis antarctica* (Cymodoceaceae) because of their phylogenetic (diverged 105 Ma ago, Waycott et al., 2018) and morphological (de los Santos et al., 2012, 2016 differences. Fresh leaves were excised and left to decompose on sediment in an ambient flow-through aquarium. Detrital photosynthesis was measured as net oxygen production of destructively sampled leaves at regular intervals over 34 days (see *Supplementary methods*). On the basis of these and previous observations (Frontier et al., 2021, 2022; Wright et al., 2022, 2024), I then developed a mathematical model which describes detrital photosynthesis as a logistic decay function of detrital age, allowing simple parameterisation of the longevity of detrital photosynthesis as photosynthetic half-life. After compiling a detrital photosynthesis dataset (Wright, 2025), I used this model in a Bayesian meta-analysis to infer the photosynthetic half-life of different plants and quantify the effects of light and choice of response variable (photosynthesis vs. chlorophyll). I then compared photosynthetic half-life with detrital half-life (sensu Olson, 1963) to estimate the relative influence of physiology on decomposition for different plants. Finally, I present a new causal model of decomposition that emerges naturally from the theory of detrital photosynthesis.

2. Methods

2.1. Model

Detached plant parts can remain in limbo for long durations, often photosynthesising at similar rates to the attached plant. For this reason and for simplicity, detrital photosynthesis has mostly been described as a linear function of detrital age (Frontier et al., 2021; Wright et al., 2022, 2024; Wright & Foggo, 2021; Wright & Kregting, 2023). But in two cases a logistic decay function was used: in a beta model predicting chlorophyll fluorescence (Fv/Fm) (Frontier et al., 2022) and in a binomial model predicting the probability of photosynthesis (Wright et al., 2024) with time after excision. The logistic function describes the decay of photosynthesis after detachment better than the linear function because it is bounded. Detrital photosynthesis cannot conceivably exceed baseline photosynthesis under identical conditions. An ill-defined baseline or changes in environment (e.g. when detritus floats and is exposed to more light) may naturally lead to brief increases after detachment (Frontier et al., 2021; Wright & Kregting, 2023). However, this cannot be considered part of the photosynthetic decay process to be modelled. Similarly, photosynthesis cannot decline beyond a set minimum, which is zero for most response variables (e.g. gross photosynthesis, chlorophyll fluorescence and chlorophyll concentration) and a negative constant for net photosynthesis. The logistic function is therefore the natural choice.

There are various ways to parameterise the logistic function. Typically, a simple linear term with intercept and slope is inserted into the standard logistic function. But for logistic decay with time (t), the intercept is not a meaningful parameter because $f(t = 0)$ is the known baseline. The slope is more meaningful because it provides the logistic rate of decay per unit t, but ultimately the parameter of interest is the sigmoid inflexion point or midpoint: t when $f(t) = 0.5$. It has the same units as t and is the only parameter that gives a directly interpretable measure of decay. As such it is analogous to the half-life $t_{½}$: t when $f(t) = 0.5$ for the exponential decay function (Olson, 1963). The logistic midpoint can be calculated as intercept divided by slope after estimating these parameters separately. But this usually results in enormous uncertainty and difficulty in calculating meaningful central tendencies (Wright et al., 2024), so the midpoint is best directly estimated. Although the slope and midpoint parameterisation may sometimes be used, these parameters are rarely identifiable. Moreover, the baseline is known to be 1 in any proportional decay model, so it is sensible to fix the intercept. Since the standard logistic function can only approach 1, a logistic intercept of 5, translating to $f(0) = 0.99$, is a reasonable approximation.

The most complex case to model is that of net photosynthesis since it has a negative lower bound. Two scaling parameters outside the standard logistic function are required. Thus my full logistic function describing the decay of photosynthesis (p) with time after detachment (t) is

$$p(t) = \frac{\alpha + \tau}{1 + e^{\frac{5(t-\mu)}{\mu}}} - \tau \quad (1)$$

where α is baseline photosynthesis, μ is the midpoint of photosynthetic decay, which I term the photosynthetic half-life, and τ is baseline respiration. $\tau = 0$ when the lower bound is zero (e.g. gross photosynthesis, chlorophyll fluorescence and chlorophyll concentration) and $\alpha = 1$ when photosynthesis is normalised relative to the baseline. The naming of parameters is inspired by the word אמת (graecised $\tau\mu\alpha$), which is composed of the first, middle and last letters of the Hebrew alphabet, symbolising life (א), transition (מ) and death (ת). In the story of the golem (McElreath, 2020), the word lends life but removing Xאמת changes its meaning from truth to death (תמ) and takes life.

2.2. Literature review

Following the seagrass experiment (see *Supplementary methods*), I searched the literature using keyword strings such as '(photosynthesis OR chlorophyll) AND ("induced senescence" OR detrit* OR detach* OR excis* OR cut OR isolate*) AND (week* OR day* OR hour* OR minute* OR time OR age)' in Web of Science (webofscience.com/wos) and Google Scholar (scholar.google.com). Once I had a selection of suitable papers, I used Crossref Metadata Search (search.crossref.org) to find similar papers. I accepted studies on all plants (sensu Bolton, 2016), the condition being that senescence was induced by excision at an exact time and photosynthesis was repeatedly measured in a timeseries of at least three timepoints over any interval to assess viability of the detached tissue. Studies on abscission were not considered since senescence starts at an unknown time prior to detachment. I naturally did not include plants that always live detached, such as pleustophytes and the holopelagic seaweeds *Sargassum fluitans* and *S. natans*, for which detrital age cannot be defined as time since detachment. I accepted various response variables, including O_2 and CO_2 gas exchange, ^{14}C assimilation, Fv/Fm, photosystem I and II electron transport rates (Mehler and Hill reactions), RuBisCo activity or concentration, carbonic anhydrase activity and photorespiration (glycolate concentration), as well as total chlorophyll or chlorophyll *a* concentration or a chlorophyll indicator. Microbial photosynthesis was generally assumed to be negligible since most of the mentioned response variables would likely not detect it. Including the present study, this resulted in 129 suitable studies between 1957 and 2025 (see *Supplementary references*) and 551 experiments (median = 3 per study).

Data were extracted by copying tables or digitising figures using WebPlotDigitizer v4.6 (Rohatgi, 2022). In two cases I contacted authors for their data, and in five cases the data were my own. Raw data were otherwise not available. The resulting dataset (Wright, 2025) contains data on 92 species from 23 orders in four non-taxonomic groups of interest: terrestrial plants (66 species), freshwater plants (6 species), seagrasses (5 species) and seaweeds (15 species). It is noteworthy that all freshwater plants in the dataset belong to the order Alismatales, like seagrasses, and all seaweeds belong to the phyla Chlorophyta and Heterokontophyta (Supplementary Table S1). The assigned taxonomy follows Plants of the World Online (powo.science.kew.org, POWO, 2025) and AlgaeBase (algaebase.org, Guiry & Guiry, 2025). My distinction between freshwater and terrestrial plants is based on whether leaves are submerged and photosynthesis can therefore be considered aquatic. There were no studies on freshwater macroalgae, so seaweeds are assumed to be representative. Several predictor variables including light and water availability as well as the type of photosynthesis measurement were also recorded. When individual observations could not be extracted, the mean, standard error of the mean (SEM) and sample size (n) were recorded instead. Uncertainty was mostly given as SEM, but when standard deviation (SD) was provided instead, it was converted as SEM = $\frac{SD}{\sqrt{n}}$. In a few cases uncertainty was given as a 95% confidence interval (CI) or interquartile range (IQR). In the former case, I converted as SEM = $\frac{0.5 \times CI_{0.95}}{\Phi^{-1}(0.975)}$, in the latter I assumed mean = median and

conservatively converted the larger of the two quartile ranges (QR: $Q_3 - Q_2$ or $Q_2 - Q_1$) as SEM = $\frac{QR}{\Phi^{-1}(0.75) \times \sqrt{n}}$.

2.3. Meta-analysis

As is typical of meta-analyses, the data contain a mixture of observations and means. To jointly analyse data with different levels of uncertainty one must either summarise observations and build a measurement error model or expand means to observations by simulation. I opted for the latter approach because in the majority of cases observations were available, and I obtained a total of 10,646 observations by simulating draws from the sampling distribution. For most response variables, which are bounded by zero, I chose the gamma distribution; for net photosynthesis I chose the normal distribution and for Fv/Fm I chose a truncated normal distribution because some summary statistics violated the beta distribution. Using the sample size (n), mean and standard deviation (sd), I implemented sampling in R v4.5.2 (R Core Team, 2026) with 'rgamma(n, mean^2/sd^2, mean/sd^2)', 'rnorm(n, mean, sd)' and 'rtnorm(n, mean, sd, 0, 1)', the last function being from extraDistr v1.10.0 (Wolodzko, 2023).

The response variable is an amalgam of various measures of photosynthesis. I therefore normalised it to enable formal comparison. Observations were expressed relative to the initial observation or mean of initial observations in the timeseries. The resulting ratios, representing relative photosynthesis, are the least invasive form of normalisation and frequently used to express photosynthetic decay (29 of 129 studies, e.g. Thimann & Satler, 1979, Jana & Choudhuri, 1980, Pellikaan, 1982). The drawback of this approach is that data are not fully brought onto the same scale because net photosynthesis allows zeros and negative ratios. I resolved this issue by replacing all zeros and negative ratios with the minimum observed positive ratio, assumed to be the minimum observable ratio within measurement error of zero. This means equating net respiration with absence of photosynthesis, which is of course not the case because respiration can mask photosynthesis in net photosynthesis measurements. Nonetheless, I believe this to be acceptable for the purposes of determining the timescales of photosynthetic decay. This normalisation allows using the simplest form of Equation (1) with $\tau = 0$ and $\alpha = 1$.

Apart from t (days), the main predictor variables of interest are plant group (G = terrestrial plants, freshwater plants, seagrasses, seaweeds), light availability (light or dark), and the type of response variable (chlorophyll or photosynthesis). Light is naturally a key predictor of photosynthetic persistence (Frontier et al., 2021, 2022; Wright et al., 2024), and the type of response variable is important because chlorophyll and photosynthesis are expected to senesce on different timescales. Of the recorded predictors, water availability is only relevant to terrestrial plants and was therefore not considered in the model. Data coverage is incomplete across the main predictor variables, with data missing for the combinations freshwater–light–photosynthesis, seagrass–dark–photosynthesis and seaweed–dark–chlorophyll. Since light availability and the type of response variable are binary predictors, they were coded as indicators of light (L = −0.5 for dark, +0.5 for light) and chlorophyll (C = −0.5 for photosynthesis, +0.5 for chlorophyll), centred on zero to avoid the bias in prior variance typical of indicator variables (McElreath, 2020) and to allow easy prediction of the mean. To get a sense of interspecific and experimental variation, I also included plant species (S, 92 levels) and experiment (E, 551 levels) as categorical predictors. The resulting Bayesian log-normal multilevel model of relative photosynthesis (p) in standard statistical notation (sensu McElreath, 2020) is

$$
\begin{aligned}
\ln p_i &\sim \text{Normal}\left(\overline{p}_i, \sigma\right) \\
\overline{p}_i &= \ln \frac{1}{1 + e^{\frac{5(t_i - \mu_i)}{\mu_i}}} \\
\ln \mu_i &= \alpha_{G_i} + \alpha_{S_i} + \alpha_{E_i} + \beta_{G_i} L_i + \gamma_{G_i} C_i \\
\alpha_G &\sim \text{Normal}\left(\overline{\alpha}, \sigma_{\boldsymbol{\alpha_G}}\right) \\
\alpha_S &\sim \text{Normal}\left(\mathbf{0}, \sigma_{\boldsymbol{\alpha_S}}\right) \\
\alpha_E &\sim \text{Normal}\left(\mathbf{0}, \sigma_{\boldsymbol{\alpha_E}}\right) \\
\beta_G &\sim \text{Normal}\left(\overline{\beta}, \sigma_\beta\right) \\
\gamma_G &\sim \text{Normal}\left(\overline{\gamma}, \sigma_\gamma\right) \\
\overline{\alpha} &\sim \text{Normal}\left(\ln \mathbf{14}, 0.3\right) \\
\overline{\beta}, \overline{\gamma} &\sim \text{Normal}\left(0, 0.4\right) \\
\sigma_{\boldsymbol{\alpha_G}}, \sigma_{\boldsymbol{\alpha_S}}, \sigma_{\boldsymbol{\alpha_E}} &\sim \text{Normal}_{0,\infty}\left(0, 0.3\right) \\
\sigma_\beta, \sigma_\gamma &\sim \text{Normal}_{0,\infty}\left(0, 0.4\right) \\
\sigma &\sim \text{Exponential}\left(1\right)
\end{aligned}
\tag{2}
$$

where the logarithm of the photosynthetic half-life μ (Equation 1) is described by a linear model with intercepts (α) indexed by G, S and E along with the effects of light (β) and measuring chlorophyll instead of photosynthesis (γ) indexed by G. β and γ are differences on the logarithmic scale, so represent log ratios. All parameters are partially pooled to enable prediction for new categories (e.g. experiments, species, plant groups), borrow information and regularise spurious effects (McElreath, 2020). The intercept standard deviations (σ_α) give an indication of intergroup, interspecific and experimental variation. The prior for the mean of μ on the logarithmic scale ($\overline{\alpha}$) is centred on ln14 because two weeks seems like a reasonable median midpoint of photosynthetic decay based on my observations on kelps (Wright et al., 2022, 2024) and literature review. Even though I suspected β and γ to be positive, I centred the priors for $\overline{\beta}$ and $\overline{\gamma}$ on zero to avoid biasing the effect.

The model was written in Stan (Carpenter et al., 2017) and implemented with CmdStan v2.17.1 (Lee et al., 2026) in R. This was facilitated by cmdstanr v0.9.0 (Gabry et al., 2025), tidybayes v3.0.7 (Kay, 2024) and bayesplot v1.14.0 (Gabry et al., 2019). Posterior probability distributions for each parameter were estimated based on 8×10^4 Markov chain Monte Carlo samples (8 chains × 10^4 iterations). Model performance was assessed using $\widehat{R}$ scores and trace rank and pair plots (Gabry et al., 2019). For data wrangling, summary and visualisation I used tidyverse v2.0.0 (Wickham et al., 2019), here v1.0.2 (Müller, 2025), ggridges v0.5.7 (Wilke, 2021), ggh4x v0.3.1 (van den Brand, 2025), geomtextpath v0.2.0 (Cameron & van den Brand, 2022), patchwork v1.3.2 (Pedersen, 2025) and officer v0.7.0 (Gohel et al., 2025). For details, see data and code at github.com/lukaseamus/plant-limbo and doi.org/10.5281/zenodo.23182864. Vector illustrations were made and added to figures in Affinity Designer v1.10.8 (Serif Ltd., Nottingham, UK).

To contextualise the photosynthetic half-life μ (days), I formally compared it with the half-life $t_{½}$ (days), its equivalent in the exponential model of decomposition. To facilitate comparison, I hereinafter refer to $t_{½}$ as the detrital half-life. $t_{½}$ is related to the exponential decay constant k by $t_{½} = \ln 2/k$ (Olson, 1963). k is usually estimated using the exponential decay function but can also be approximated by the detrital turnover rate P/B, the ratio of detrital production to the detrital pool

Fig. 1.- Colour online, B/W in print

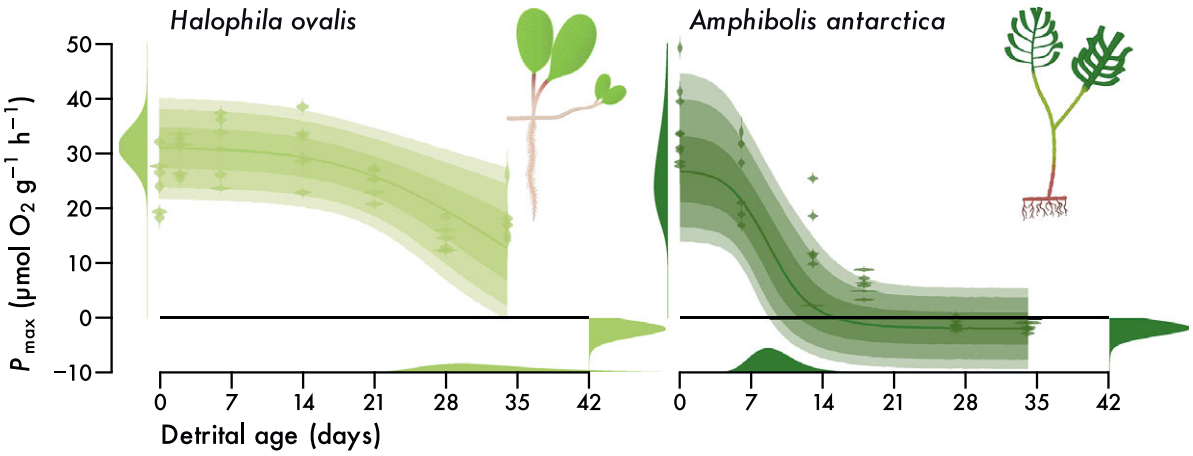


Figure 1. Seagrass detrital photosynthesis. Light-saturated net photosynthesis (P_{max}) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves over 34 days post-excision. P_{max} is standardised by leaf fresh mass. Violins are kernel density estimates of observations with measurement error (n = 42 for *H. ovalis*, n = 39 for *A. antarctica*). Lines and intervals are medians and the central 50%, 80% and 90% of posterior probability for predicted observations. Distributions are posteriors for initial P_{max} (α), photosynthetic half-life (μ) and final respiration (τ) correctly mapped onto the axes (cf. Equation 1).

(Cebrián & Lartigue, 2004). Using published datasets (Anderson-Teixeira et al., 2018; Cebrián & Lartigue, 2004; Filbee-Dexter et al., 2024; Kennedy & Blain, 2025; White & Norkko, 2025; Wright, 2026a), I thus extracted or calculated k for terrestrial plants, freshwater plants, seagrasses and seaweeds, and converted it to $t_{½}$. I then modelled $t_{½}$ with a log-normal multilevel intercept model (see code for details) and compared its predictions to those for μ from Equation (2)). I formalised this comparison as the ratio $\mu/t_{½}$, a measure of physiological influence on decomposition. $\mu/t_{½} = 1$ implies that detrital photosynthesis reaches half its initial value at the same time as detrital mass. If $\mu/t_{½} < 1$, photosynthesis decays faster than mass and is unlikely to influence decomposition. $\mu/t_{½} > 1$ is theoretically impossible because there can be no photosynthesis without mass, but in practice it will occur due to the difference between logistic and exponential decay trajectories and empirical uncertainty. $\mu/t_{½} > 1$ may be interpreted as a strong indication that photosynthesis determines decomposition. $\mu/t_{½}$ was calculated for each plant group from the independently estimated posterior probability distributions for μ and $t_{½}$. This approach propagates all uncertainty and allows calculation of the probability that the ratio falls above or below 1.

3. Results

3.1. Seagrasses

H. ovalis photosynthesised for longer post-excision than *Amphibolis antarctica*. After 34 days, the duration of the experiment, all excised *H. ovalis* leaves were still obviously net autotrophic while all excised *A. antarctica* leaves were apparently dead. On a mass basis, *H. ovalis* displayed similar baseline light-saturated net photosynthesis (P_{max}) to *A. antarctica*, but its photosynthetic half-life (μ in Equation 1) was 3.9 ± 1.6 (mean ± standard deviation) times as long (Figure 1, Table 1). On a leaf basis, baseline P_{max} of *A. antarctica* was naturally much greater but the photosynthetic half-life of *H. ovalis* was 6.8 ± 2.5 times as long (Table 1).

Mean leaf blotted mass of *H. ovalis* (initially 0.052 ± 0.005 g $leaf^{-1}$) declined at linear rates of 0.58 ± 0.26 mg $leaf^{-1}$ day^{-1} or 1.1 ± 0.43% day^{-1} while that of *A. antarctica* (initially 0.64 ± 0.053 g $leaf^{-1}$) only declined at 2 ± 2.7 mg $leaf^{-1}$ day^{-1} or 0.3 ± 0.41% day^{-1}. I also observed periphyton colonising *H. ovalis* but not *A. antarctica*. These observations cast some doubt on the apparent greater photosynthetic half-life of excised *H. ovalis* leaves. If periphyton contributed to photosynthesis, it may have exacerbated the observed difference in photosynthetic persistence between the two seagrasses. Accounting for the uncertainty introduced by these limitations, I expect seagrasses to have a photosynthetic half-life of more than a week (*A. antarctica*) but less than a month (*H. ovalis*).

3.2. Meta-analysis

Comparison of detrital photosynthesis across various plants revealed that light ameliorates photosynthetic decay. Detritus of terrestrial plants, seagrasses and seaweeds with access to light is predicted to have a photosynthetic half-life that is 1.8, 1.6 and 2 times as long as that in darkness (Figure 2, Table 2). Freshwater plants form an exception with a mere 68% chance (only somewhat above 50:50) of extended photosynthesis under light (Figure 2, Table 2). The predicted median light–dark ratio for unobserved plant groups is therefore 1.5 (ln ratio = 0.4 ± 0.52). In absolute terms, however, the light effect is only substantial for seaweeds (Figure 2). While median terrestrial photosynthetic half-life is only expected to increase from 7.2 to 13 days with access to light, that of seaweeds increases from 24 to 51 days (Supplementary Table S2). In darkness, seaweed detritus thus only remains viable for comparable durations to other plants (Figure 2; Supplementary Table S2).

The choice of response variable also matters when estimating detrital photosynthesis. Chlorophyll generally decays slower than photosynthesis. This pattern is consistent across all plants with photosynthetic half-life predicted to be 1.4 times (ln ratio = 0.33 ± 0.36) as long for chlorophyll as for photosynthesis (Figure 2, Table 2). In absolute terms, the effect again dominates for seaweeds (Figure 2), with an increase from 51 days for photosynthesis to 81 days for chlorophyll under light (Supplementary Table S2). Measurements based on chlorophyll therefore tend to overestimate the longevity of detrital photosynthesis.

Across light conditions and response variables, the median half-life of detrital photosynthesis ranges from 11 days for terrestrial plants, over 14 and 17 days for freshwater plants and seagrasses to 44 days for seaweeds (Table 2). The prediction for unobserved plant groups is 17 days (ln = 2.8 ± 1.1). In relative terms, seaweed

Table 1. Seagrass parameter estimates (mean ± standard deviation) for initial light-saturated net photosynthesis (α), photosynthetic half-life (μ) and final respiration (τ)

Parameter	*Halophila ovalis*	*Amphibolis antarctica*	Δ	$P(\Delta > 0)$
Mass-based				
α (μmol g^{-1} h^{-1})	31 ± 3.4	28 ± 8.4	3.3 ± 9.1	0.69
μ (days)	35 ± 11	9.5 ± 2.6	26 ± 11	1
τ (μmol g^{-1} h^{-1})	2.1 ± 0.85	1.9 ± 0.76	0.17 ± 1.1	0.56
Leaf-based				
α (μmol $leaf^{-1}$ h^{-1})	1.6 ± 0.17	33 ± 12	31 ± 12	1
μ (days)	46 ± 15	7 ± 1.3	39 ± 15	1
τ (μmol $leaf^{-1}$ h^{-1})	0.92 ± 0.38	0.93 ± 0.37	0.0049 ± 0.53	0.51

Note: Δ is the absolute difference between estimates for H. ovalis (n = 42) and A. antarctica (n = 39). $P(\Delta > 0)$ is the probability that estimates for H. ovalis and A. antarctica are different. P = 0.5 is equivalent to a coin toss.

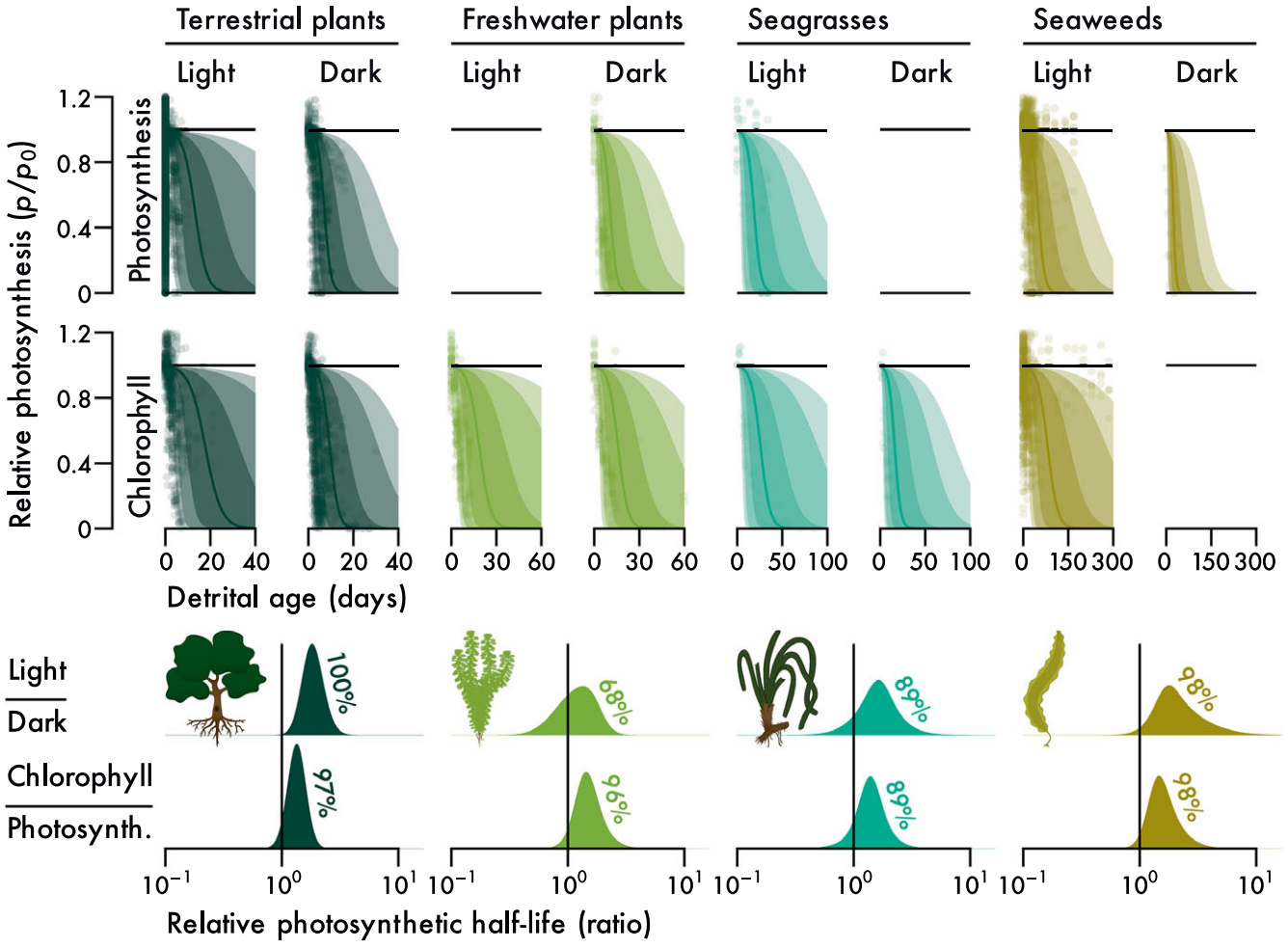


Figure 2. The effect of light and response variable type on detrital photosynthesis for major plant groups: terrestrial plants (Streptophyta excluding aquatic plants), freshwater plants (Alismatales excluding seagrasses), seagrasses, and seaweeds (Chlorophyta and Heterokontophyta excluding freshwater macroalgae). Points are observations (n = 7,240, 673, 160 and 2,573 for terrestrial plants, freshwater plants, seagrasses and seaweeds). Note that there are several observations outside the axes ranges. Lines and intervals are medians and the central 50, 80 and 90% of posterior probability for the median prediction ($e^{\bar{\beta}}$ in Equation 2). Note that data are missing where panels are empty, and predictions are therefore not shown. Distributions are kernel density estimates of posterior probabilities for the light and chlorophyll relative effect sizes (e^{β} and e^{γ} in Equation 2). Percentages are probabilities for $\beta > 0$ and $\gamma > 0$.

detrital photosynthesis persists 3.9, 3.1 and 2.5 times as long as that of terrestrial plants, freshwater plants and seagrasses (Table 3; Supplementary Figure S2). All vascular plant detritus remains viable for similar durations but there is a tendency for that of freshwater plants to have a somewhat longer photosynthetic half-life compared to terrestrial plants, with that of seagrasses being longer still (Table 3, Supplementary Figure S2).

Notwithstanding these general trends, there is substantial interspecific and experimental variation in photosynthetic half-life (Figure 3). Median estimates for species range from 3–4 days

Table 2. Meta-analysis parameter estimates (Equation 2) for plant groups

Parameter	Terrestrial plants	Freshwater plants	Seagrasses	Seaweeds
n	7,240	673	160	2,573
μ (days)	11 (2.4 ± 0.92)	14 (2.7 ± 0.95)	17 (2.9 ± 0.97)	44 (3.8 ± 0.96)
e^{β}	1.8 (0.61 ± 0.18)	1.2 (0.16 ± 0.37)	1.6 (0.47 ± 0.4)	2 (0.77 ± 0.47)
$P(\beta > 0)$	1	0.68	0.89	0.98
e^{γ}	1.3 (0.29 ± 0.15)	1.5 (0.41 ± 0.26)	1.4 (0.32 ± 0.3)	1.5 (0.47 ± 0.29)
$P(\gamma > 0)$	0.97	0.96	0.89	0.98
$t_{½}$ (days)	409 (6 ± 1.5)	57 (4 ± 1.5)	68 (4.2 ± 1.5)	14 (2.7 ± 1.5)
$\mu/t_{½}$	0.028 (−3.6 ± 1.8)	0.25 (−1.4 ± 1.8)	0.25 (−1.4 ± 1.8)	3 (1.1 ± 1.8)
$P(\mu/t_{½} > 1)$	0.02	0.22	0.23	0.73

Note: Photosynthetic half-life (μ) is predicted for new experiments on unobserved species across the light (β) and chlorophyll (γ) effects. Exponentiating β and γ yields relative effect sizes (cf. Figure 2). μ is also expressed relative to detrital half-life ($t_{½}$) to quantify the physiological influence on decomposition (cf. Figure 3). All parameter estimates are given as median (ln mean ± standard deviation) to accurately represent right-skewed probability distributions. n and P denote sample size and probability.

Table 3. Photosynthetic (μ) and detrital ($t_{½}$) half-life contrasts between plant groups (cf. Supplementary Figure S2)

Ratio	μ/μ	$P(\mu/\mu > 1)$	$t_{½}/t_{½}$	$P(t_{½}/t_{½} > 1)$
Seaweeds/Terrestrial	3.9 (1.4 ± 1.3)	0.85	0.035 (−3.3 ± 2.1)	0.057
Seaweeds/Freshwater	3.1 (1.1 ± 1.4)	0.80	0.26 (−1.4 ± 2.1)	0.26
Seaweeds/Seagrasses	2.5 (0.93 ± 1.4)	0.75	0.21 (−1.6 ± 2.1)	0.23
Seagrasses/Terrestrial	1.5 (0.43 ± 1.3)	0.63	0.17 (−1.8 ± 2.1)	0.2
Freshwater/Terrestrial	1.3 (0.23 ± 1.3)	0.57	0.14 (−2 ± 2.1)	0.18
Seagrasses/Freshwater	1.2 (0.2 ± 1.3)	0.56	1.2 (0.18 ± 2.2)	0.53

Note: P(ratio > 1) is the probability that estimates are different. P = 0.5 is equivalent to a coin toss. All estimates except P are given as median (ln mean ± standard deviation). See Table 2 for sample sizes.

for the wheat *Triticum aestivum* to 154 days for the Antarctic brown seaweed *Desmarestia anceps* (Supplementary Table S3). Median estimates for experiments range from 1.2 to 438 days. The variance in photosynthetic half-life between plant groups, species and experiments is directly quantified by partial pooling standard deviations on the logarithmic scale (σ_{α} in Equation 2). These standard deviations respectively are 0.51 ± 0.16, 0.53 ± 0.12 and 0.72 ± 0.057 for plant groups, species and experiments, so experimental variation is the greatest source of uncertainty.

Decomposition timescales also vary substantially across plant groups. Terrestrial plant detritus decomposes very slowly, only reaching half its initial mass after over a year on average (Figure 3, Table 2). Aquatic angiosperms decompose much faster, with a median detrital half-life ranging from 57 days for freshwater plants to 68 days for seagrasses (Figure 3, Table 2), only 14% to 17% of terrestrial half-life (Table 3). Seaweeds decompose fastest, with a median detrital half-life of only two weeks (Figure 3, Table 2), 21–26% of aquatic angiosperm half-life and only 3.5% of terrestrial half-life (Table 3). Photosynthetic half-life must therefore be viewed in relation to detrital half-life to assess if decomposition is affected by detrital photosynthesis.

The ratio of photosynthetic to detrital half-life ($\mu/t_{½}$) quantifies the relative influence of detrital photosynthesis on decomposition, where ratios below 1 indicate little influence and ratios of 1 and above indicate substantial influence. The median of $\mu/t_{½}$ ranges from 0.028 for terrestrial plants to 0.25 for freshwater plants and seagrasses to 3 for seaweeds (Figure 3, Table 2). Seaweeds are therefore the only plants with a median ratio above 1, and there is a 73% chance that the ratio for any seaweed is above 1, while this probability is 22–23% for aquatic angiosperms and only 2% for terrestrial plants (Figure 3, Table 2).

4. Discussion

Some plant detritus can maintain photosynthesis for long periods following detachment. This is at odds with conventional understanding of detritus as dead organic matter. Despite the existence of various relevant literature published on this topic since the 1950s and recognition of the phenomenon in seaweeds since the 1980s, detrital photosynthesis had not been explored beyond kelps or formally conceptualised. In part, this is probably due to diverse research objectives and terminology which obscure the connection between quantitatively comparable data. There was also data scarcity for seagrasses which, as the only vascular counterpart to seaweeds, form an important functional link to freshwater and terrestrial plants. Here, I collected the first data on photosynthesis in seagrass detritus, developed a mathematical model of detrital photosynthesis and used it to make the first quantitative comparison of photosynthetic persistence in the detrital phase of various plants. Results suggest that the non-vascular macroalgae have the greatest potential for detrital photosynthesis while among the vascular plants there is slightly increased detrital viability with a shift from a terrestrial to an aquatic and yet again to a marine lifestyle. Although all plants display detrital photosynthesis to some extent, it does not necessarily influence their decomposition.

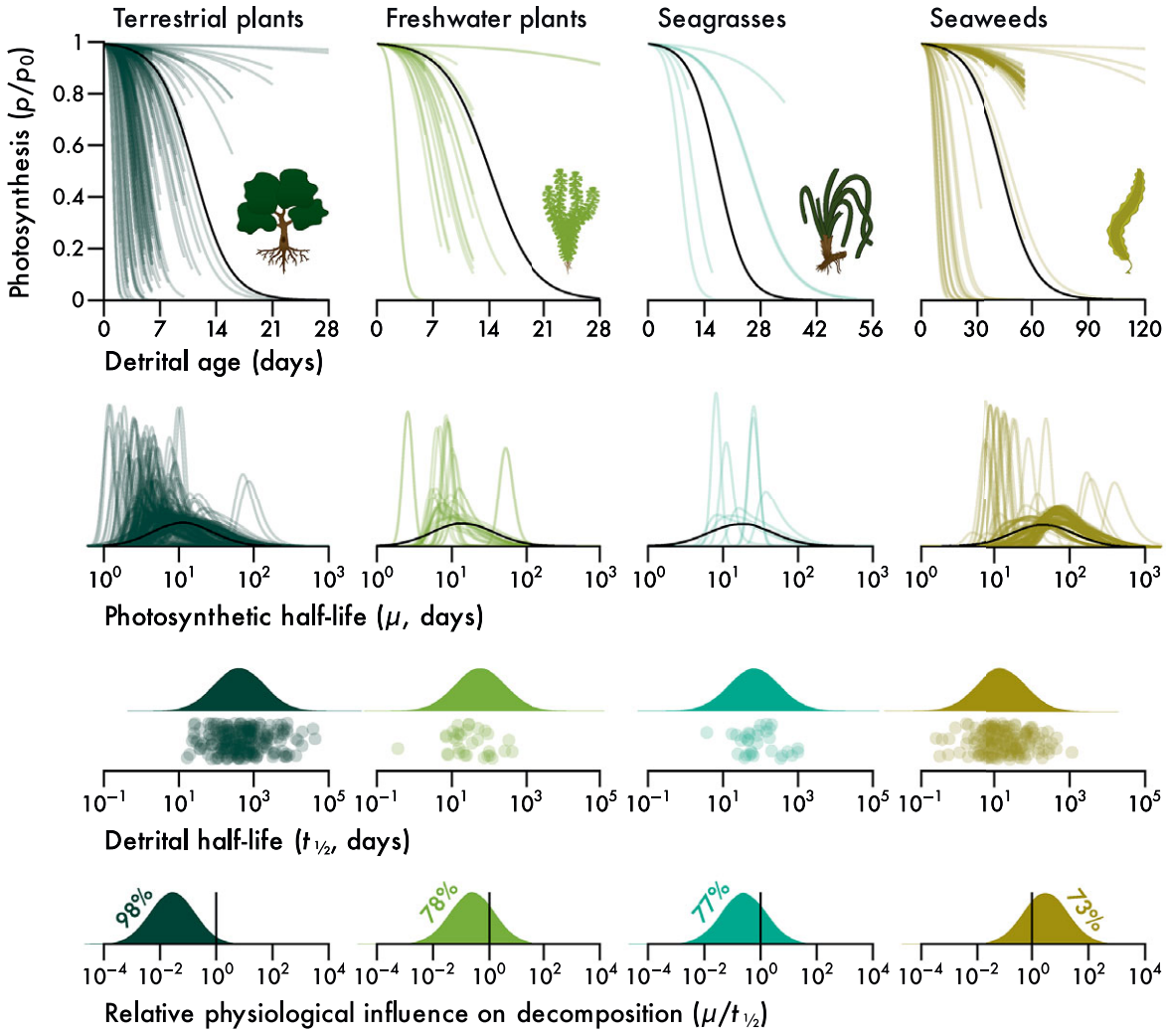


Figure 3. Longevity of detrital photosynthesis relative to detrital mass for major plant groups (cf. Figure 2). Lines are median predictions of relative photosynthesis ($e^{\bar{p}}$ in Equation 2) or posterior probability distributions of photosynthetic half-life (μ in Equation 2). Existing experiments (n = 314, 32, 8 and 197 for terrestrial plants, freshwater plants, seagrasses and seaweeds) are given in colour and predictions for new experiments on unobserved species across the light and chlorophyll effects are given in black. Raincloud plots of detrital half-life ($t_{½}$ sensu Olson, 1963) show observations (n = 239, 33, 32 and 171 for terrestrial plants, freshwater plants, seagrasses and seaweeds) below posterior probability distributions for new observations. Distributions in the bottom panel are ratios of the black distributions of μ to the distributions of $t_{½}$. Percentages are probabilities for $\mu/t_{½} < 1$ or $\mu/t_{½} > 1$, depending on what side of the identity line most of the probability lies.

Terrestrial plant decomposition is slower than that of aquatic vascular plants which in turn is much slower than that of macroalgae. I predict that detrital photosynthesis is a general phenomenon but its effect on decomposition ranges from negligible for terrestrial plants, over measurable for aquatic vascular plants, to substantial for macroalgae. It is conceivable that detrital photosynthesis determines macroalgal decomposition. I therefore propose a new causal model of macroalgal decomposition that emerges from my formalisation of the theory of detrital photosynthesis (Figure 4). While conventional decomposition theory may continue to be applied to vascular plants and detritus sensu stricto, its assumptions will not always hold for aquatic systems and certainly do not apply to macroalgae (Figure 4).

Detrital photosynthesis per se is not restricted to non-vascular plants. Seagrass detritus can also photosynthesise for at least a week. The oxygen production of the diverse seagrasses *H. ovalis* and *Amphibolis antarctica* responds similarly to detrital age, and the delayed response in *H. ovalis* may simply be attributable to greater colonisation by periphyton. Given a leaf mass per area of 2.7 and 6.8 mg cm^{-2} for *H. ovalis* and *A. antarctica* respectively (de los Santos et al., 2012; de los Santos et al., 2016, periphyton could contribute 12–172% of initial seagrass photosynthesis (Neely & Wetzel, 1997). Alternatively, *A. antarctica* leaves may be more reliant on their roots for nutrient provisioning, although there is currently no evidence to suggest this (Pedersen et al., 1997). The 26–39-day shorter photosynthetic half-life of isolated *A. antarctica* leaves compared to *H. ovalis* is surprising given their 1.7 times or 47-day longer attached lifespan (Hemminga et al., 1999). Nonetheless, *Halophila* detritus is often viable (Hall et al., 2006; Weatherall et al., 2025) and can continue growing to a similar or greater extent relative to other seagrasses (Hanuise et al., 2026). Despite observations of seagrass detrital viability (Hall et al., 2006; Stafford-Bell et al., 2015; Weatherall et al., 2025) and growth (Hanuise et al., 2026) for up to a month and my evidence of photosynthetic persistence for a similar duration, seagrass detrital photosynthesis may not necessarily counteract decomposition. I found that seagrass detritus has a detrital half-life of 68 days on average, and it may take a year for the

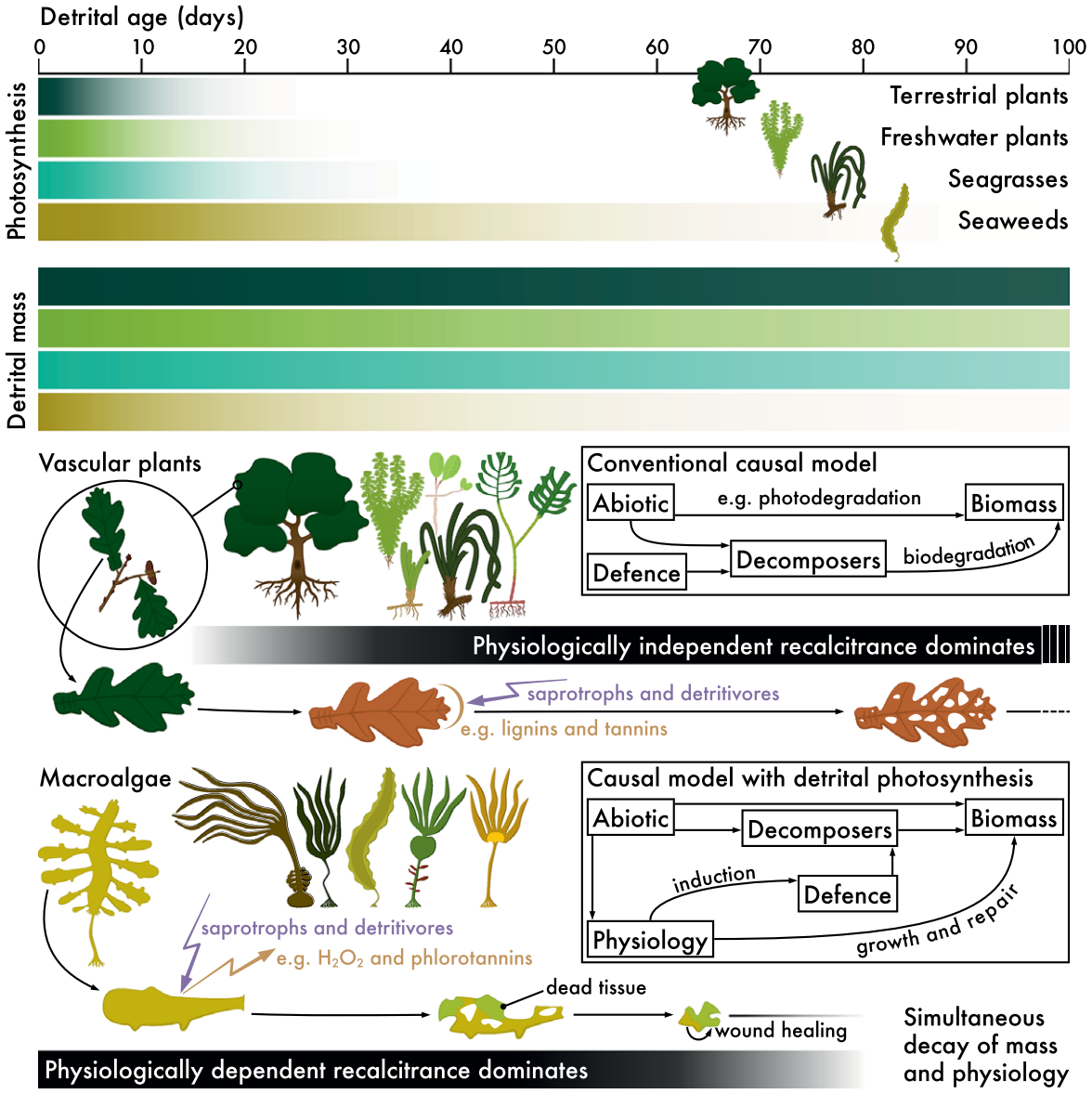


Figure 4. Conceptual model of decomposition in vascular plants and macroalgae. Coloured bars show the decay of detrital photosynthesis and biomass for major plant groups, based respectively on the median for the posterior prediction of relative photosynthesis p (Equation 2) and e^{-kt} where the decay rate k is derived from detrital half-life ($t_{½}$) as the median of $\ln 2/t_{½}$ (cf. Figure 3). Black bars show approximate timelines for the dominance of physiologically independent and dependent resistance to decomposition (i.e. recalcitrance) in vascular plants and macroalgae respectively. Insets show the conventional and new causal models as directed acyclic graphs (McElreath, 2020) where nodes are variables and arrows show their causal relationships. Some arrows are labelled with mechanisms for clarity. Illustrations connected by arrows show the schematic decomposition process over time. Note that some illustrated concepts are speculative and further empirical work is needed to validate them (see Conclusion).

majority to decompose (Cebrián et al., 1997; Cebrián & Lartigue, 2004;Harrison, 1989; Trevathan-Tackett et al., 2020). This is much longer than detached seagrass is expected to photosynthesise, and accordingly there is little evidence of physiological influence on its decomposition. While vegetative fragments of the paddle grass *Halophila decipiens* do indeed decompose faster when buried (Kenworthy et al., 1989), decomposition rates are similar between live and senescent or buried leaves of the turtle grass *Thalassia testudinum* (Fourqurean & Schrlau, 2003; Rosch & Koch, 2009; Rublee & Roman, 1982) and reviews suggest that live seagrass leaves actually tend to decompose faster than senescent or dead ones (Harrison, 1989; Trevathan-Tackett et al., 2020).

Detrital photosynthesis is favoured primarily by non-vascularity and secondarily by an aquatic lifestyle. Macroalgae uniquely exhibit long-term detrital photosynthesis, which clearly sets them apart from vascular plants (Figure 4). Nevertheless, seagrasses and freshwater plants on average display detrital photosynthesis several days longer than terrestrial plants. This is almost certainly due to the greater photosynthetic independence of aquatic plant leaves. Because most experiments are carried out in water, they probably overestimate the persistence of detrital photosynthesis in terrestrial plants. With access to light but not water, which is the usual fate of terrestrial plant detritus, detached leaves of various trees tend to lose the ability to photosynthesise in just a few minutes (e.g. Gauthier & Jacobs, 2018; Kar et al., 2021) due to breakdown of stomatal conductance following loss of water pressure (Gauthier & Jacobs, 2018; Powles et al., 2006). The true difference between aquatic and terrestrial vascular plants may therefore be greater than

predicted by my meta-analysis. Surprisingly, seagrass detritus tends to photosynthesise for longer than that of freshwater plants from the same order. The adoption of a marine lifestyle and convergent evolution with seaweeds (Olsen et al., 2016) may be the ultimate reason. But a more proximate explanation is excess production of hydrogen peroxide in detached freshwater plant leaves (Jana & Choudhuri, 1982; Kar & Choudhuri, 1986, 1987), which may be absent in seagrass leaves. As demonstrated for the water thyme *Hydrilla verticillata* (Kar & Choudhuri, 1986, 1987), hydrogen peroxide production is exacerbated by photosynthesis, leading to light-induced chlorophyll degradation that is independent of photodegradation. Hence freshwater plants were the only group for which I did not find a positive light effect on photosynthetic half-life (Figure 2). In contrast, there is no evidence for increased chlorophyll breakdown in isolated seagrass leaves exposed to light (Strother & Vatta, 1986), and seagrass photodegradation has only been reported for dead leaves (Vähätalo et al., 1998). Perhaps the answer lies in alterations to the light harvesting complexes and cell wall features of seagrass leaves that enhance gas exchange (Olsen et al., 2016). Regardless, emancipation of leaf photosynthesis from the rest of the plant is apparently even more favoured by a marine than an aquatic lifestyle.

Macroalgal decomposition is uniquely determined by detrital photosynthesis. Of all considered plants, terrestrial ones decompose slowest and macroalgae decompose fastest, so the observed divergence in photosynthetic decay is increased when viewing it relative to decomposition. Macroalgae are the only plants whose detritus may photosynthesise for as long as it remains intact; aquatic vascular plants maintain photosynthesis for a quarter of their detrital phase, and terrestrial decomposition is probably unaffected by physiology due to a mismatch between rapid photosynthetic decay and slow decomposition (Figure 4). A clear distinction must here be made between vascular plants and macroalgae. Since the majority of vascular plant decomposition occurs in the absence of photosynthesis, its main intrinsic predictor must be physiologically independent constitutive defences such as lignins and tannins, while that of macroalgal decomposition is likely physiologically induced resistance to decay (Figure 4). Powered by photosynthesis, detrital physiology may be able to respond to decomposition with growth, repair and chemical defence (Figure 4). These mechanisms can go hand in hand as in the case of phlorotannins acting both as repair and defence in excised brown seaweeds (Halm et al., 2011; Lüder & Clayton, 2004). While there is evidence for the persistence and increase of detrital phlorotannins (Tala et al., 2013, 2016, Tala et al., 2019, de Bettignies et al., 2020, Wright et al., 2022, but see Pedersen et al., 2021), other common chemical defences such as induced (Küpper et al., 2001, 2002; Thomas et al., 2014; Weinberger et al., 1999; Weinberger & Friedlander, 2000) and constitutive (Küpper et al., 2002; Twigg et al., 2020) hydrogen peroxide release have yet to be measured in detritus. Nonetheless, the presented evidence for detrital photosynthesis clearly introduces novel causal paths (Figure 4). For example, light prolongs detrital physiology, which slows macroalgal decomposition, providing a mechanism contrary to photodegradation. In this case the mechanism is clear, but different mechanisms can also produce the same outcome (Figure 4). For instance, elevated temperature may accelerate decomposition in three ways: by reducing the structural integrity of detritus, by increasing decomposer metabolism, and also by diminishing detrital photosynthesis as was recently demonstrated (Wright et al., 2024). Further research is required to tease apart these causal relationships.

Detrital photosynthesis should be considered when making inferences about macroalgal decomposition. Firstly, the initial physiological state of detritus in a decomposition experiment may not reflect that in nature. In contrast to seagrass litter bag experiments (Harrison, 1989; Trevathan-Tackett et al., 2020), most seaweed decomposition experiments have only used freshly excised blades as a detritus proxy (Filbee-Dexter et al., 2022; Frontier et al., 2021, 2022; Kennedy & Blain, 2025; White & Norkko, 2025; Wright et al., 2022). Senescent or naturally detached fronds are rarely included (de Bettignies et al., 2020; Hamersley et al., 2015) and the practice of pre-killing tissue to obtain detritus sensu stricto is limited to few, mostly early studies (Birch et al., 1983; Brouwer, 1996; Josselyn & Mathieson, 1980; Rice & Tenore, 1981; Smith & Foreman, 1984). While the fraction of seaweed detritus that consists of whole dislodged plants or fronds is certainly initially viable, the majority of tissue that is abscised or eroded likely already exhibits impaired photosynthesis before detachment (Arnold & Manley, 1985; Gómez et al., 1996; Küppers & Kremer, 1978; Rodriguez et al., 2016). Ignorance of the physiology of experimental tissues on the one hand leads to the erroneous conclusion that decomposition rates derived from freshly excised seaweeds are representative of global carbon cycling (e.g. Filbee-Dexter et al., 2024), and on the other hand results in practices that inadvertently impair photosynthesis to an unknown degree, such as long sample transport (e.g. Pedersen et al., 2021). Secondly, conditions during a decomposition experiment can have unintended effects on detrital photosynthesis which may then be misattributed. As I have shown, detrital photosynthesis is highly sensitive to light in most plants (Figure 2) and temperature in kelps (Wright et al., 2024). Oxygen is undoubtedly also important, and there are probably many more nuanced physiological stressors that remain overlooked. For example, decomposition rates measured under any influence of even weak light will be highly misleading when the goal is to predict carbon cycling in the deep ocean (Filbee-Dexter et al., 2024; White & Norkko, 2025). Or the accelerating effect of temperature on decomposition may be attributed to structural weakening of detritus and elevated microbial metabolism rather than impaired detrital photosynthesis (Filbee-Dexter et al., 2022; Frontier et al., 2022). These confounders are not immediately obvious without the causal link provided by detrital photosynthesis (Figure 4). Detrital physiology likely accounts for the large unexplained variation in macroalgal decomposition rates observed across experiments (Kennedy & Blain, 2025; White & Norkko, 2025). Awareness of the causation revealed by the presented theory (Figure 4) is necessary to properly design and interpret decomposition experiments involving macroalgae and perhaps aquatic plants in general.

To conclude, I show that all plants may photosynthesise following detachment, but timescales vary greatly from a few minutes in tree leaves without access to water to a year for certain seaweeds. Only macroalgal detrital photosynthesis decays on similar timescales to detrital mass. Physiology is therefore expected to substantially influence the decomposition of macroalgae, a causal relationship which may be unique among plants. The initial physiological state of detritus and predictors of photosynthesis, such as light and temperature, can thus aid inference of macroalgal decomposition. With the theory of detrital photosynthesis formalised, we now need (1) a mathematical model of macroalgal decomposition that takes this theory into account – I have recently made some progress in this direction (Wright, 2026b), (2) more decomposition experiments comparing live with dead detritus (cf. Birch et al., 1983; Brouwer, 1996) or fresh with senescent tissues

(cf. de Bettignies et al., 2020; Hamersley et al., 2015) to corroborate the proposed effect of physiology on decomposition, (3) surveys of the initial physiological state of detritus produced by erosion, abscission and dislodgement, (4) in situ measurements of detrital photosynthesis in naturally deposited detrital accumulations, (5) further studies on post-excision photosynthesis, especially in underrepresented plants such as seagrasses and seaweeds other than kelps, and for missing treatment combinations such as freshwater plants in light and seagrasses in darkness (Figure 2) and (6) investigation into other aspects of detrital physiology such as induced chemical defence.

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Competing interest. I declare no conflicts of interest.

Data availability statement. Data and annotated code are available at github.com/lukaseamus/plant-limbo and doi.org/10.5281/zenodo.23182864. The dataset used for meta-analysis (Wright, 2025) can also be accessed at github.com/lukaseamus/detrital-photosynthesis.

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References

Anderson-Teixeira, K. J., **Wang, M. M. H.**, **McGarvey, J. C.**, **Herrmann, V.**, **Tepley, A. J.**, **Bond-Lamberty, B.**, & **LeBauer, D. S.** (2018). ForC: A global database of forest carbon stocks and fluxes. *Ecology*, **99**(6), 1507. https://doi.org/10.1002/ecy.2229

Arnold, K. E., & **Manley, S. L.** (1985). Carbon allocation in *Macrocystis pyrifera* (Phaeophyta): Intrinsic variability in photosynthesis and respiration. *Journal of Phycology*, **21**(1), 154–167. https://doi.org/10.1111/j.0022-3646.1985.00154.x

Birch, P. B., **Gabrielson, J. O.**, & **Hamel, K. S.** (1983). Decomposition of *Cladophora*. I. Field studies in the Peel-Harvey estuarine system, Western Australia. *Botanica Marina*, **26**(4), 165–172. https://doi.org/10.1515/botm.1983.26.4.165

Bolton, J. J. (2016). What is aquatic botany? – And why algae are plants: The importance of non-taxonomic terms for groups of organisms. *Aquatic Botany*, **132**, 1–4. https://doi.org/10.1016/j.aquabot.2016.02.006

Bourgueès, S., **Auby, I.**, **de Wit, R.**, & **Labourg, P.-J.** (1996). Differential anaerobic decomposition of seagrass (*Zostera noltii*) and macroalgal (*Monostroma obscurum*) biomass from Arcachon Bay (France). *Hydrobiologia*, **329**(1-3), 121–131. https://doi.org/10.1007/bf00034552

Brouwer, P. E. M. (1996). Decomposition in situ of the sublittoral Antarctic macroalga *Desmarestia anceps* Montagne. *Polar Biology*, **16**(2), 129–137. https://doi.org/10.1007/bf02390433

Cameron, A., & **van den Brand, T.** (2022). *geomtextpath: use curved text in ggplot2*. https://allancameron.github.io/geomtextpath

Carpenter, B., **Gelman, A.**, **Hoffman, M. D.**, **Lee, D.**, **Goodrich, B.**, **Betancourt, M.**, **Brubaker, M.**, **Guo, J.**, **Li, P.**, & **Riddell, A.** (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, **76**, 1–32. https://doi.org/10.18637/jss.v076.i01

Cebrián, J., **Duarte, C. M.**, **Marbà, N.**, & **Enríquez, S.** (1997). Magnitude and fate of the production of four co-occurring Western Mediterranean seagrass species. *Marine Ecology Progress Series*, **155**, 29–44. https://doi.org/10.3354/meps155029

Cebrián, J., & **Lartigue, J.** (2004). Patterns of herbivory and decomposition in aquatic and terrestrial ecosystems. *Ecological Monographs*, **74**(2), 237–259. https://doi.org/10.1890/03-4019

de Bettignies, F., **Dauby, P.**, **Thomas, F.**, **Gobet, A.**, **Delage, L.**, **Bohner, O.**, **Loisel, S.**, & **Davoult, D.** (2020). Degradation dynamics and processes associated with the accumulation of *Laminaria hyperborea* (Phaeophyceae) kelp fragments: An in situ experimental approach. *Journal of Phycology*, **56**(6), 1481–1492. https://doi.org/10.1111/jpy.13041

de los Santos, C. B., **Brun, F. G.**, **Onoda, Y.**, **Cambridge, M. L.**, **Bouma, T. J.**, **Vergara, J. J.**, & **Pérez-Lloréns, J. L.** (2012). Leaf-fracture properties correlated with nutritional traits in nine Australian seagrass species: Implications for susceptibility to herbivory. *Marine Ecology Progress Series*, **458**, 89–102. https://doi.org/10.3354/meps09757

de los Santos, C. B., **Onoda, Y.**, **Vergara, J. J.**, **Pérez-Lloréns, J. L.**, **Bouma, T. J.**, **La Nafie, Y. A.**, **Cambridge, M. L.**, **Brun, F. G.** (2016) A comprehensive analysis of mechanical and morphological traits in temperate and tropical seagrass species. *Marine Ecology Progress Series* **551**, 81–94. https://doi.org/10.3354/meps11717.

Drew, E. A. (1983). Physiology of *laminaria*. I. Use of excised lamina discs in short and long term experiments. *Marine Ecology*, **226**, 259–270. https://doi.org/10.1111/j.1439-0485.1983.tb00297.x

Evrard, V., **Kiswara, W.**, **Bouma, T. J.**, & **Middelburg, J. J.** (2005). Nutrient dynamics of seagrass ecosystems: ^{15}N evidence for the importance of particulate organic matter and root systems. *Marine Ecology Progress Series*, **295**, 49–55. https://doi.org/10.3354/meps295049

Filbee-Dexter, K., **Feehan, C. J.**, **Smale, D. A.**, **Krumhansl, K. A.**, **Augustine, S.**, **de Bettignies, F.**, **Burrows, M. T.**, **Byrnes, J. E. K.**, **Campbell, J.**, **Davoult, D.**, **Dunton, K. H.**, **Franco, J. N.**, **Garrido, I.**, **Grace, S. P.**, **Hancke, K.**, **Johnson, L. E.**, **Konar, B.**, **Moore, P. J.**, **Norderhaug, K. M.**, . . . **Wernberg, T.** (2022). Kelp carbon sink potential decreases with warming due to accelerating decomposition. *PLoS Biology*, **20**(8), e3001702. https://doi.org/10.1371/journal.pbio.3001702

Filbee-Dexter, K., **Pessarrodona, A.**, **Pedersen, M. F.**, **Wernberg, T.**, **Duarte, C. M.**, **Assis, J.**, **Bekkby, T.**, **Burrows, M. T.**, **Carlson, D. F.**, **Gattuso, J.-P.**, **Gundersen, H.**, **Hancke, K.**, **Krumhansl, K. A.**, **Kuwae, T.**, **Middelburg, J. J.**, **Moore, P. J.**, **Queirós, A. M.**, **Smale, D. A.**, **Sousa-Pinto, I.**, . . . **Krause-Jensen, D.** (2024). Carbon export from seaweed forests to deep ocean sinks. *Nature Geoscience*, **17**(6), 552–559. https://doi.org/10.1038/s41561-024-01449-7

Fourqurean, J. W., & **Schrlau, J. E.** (2003). Changes in nutrient content and stable isotope ratios of C and N during decomposition of seagrasses and mangrove leaves along a nutrient availability gradient in Florida bay, USA. *Chemistry and Ecology*, **19**(5), 373–390. https://doi.org/10.1080/02757540310001609370

Fraser, C. I., **Morrison, A. K.**, **Hogg, A. M.**, **Macaya, E. C.**, **van Sebille, E.**, **Ryan, P. G.**, **Padovan, A.**, **Jack, C.**, **Valdivia, N.**, & **Waters, J. M.** (2018). Antarctica's ecological isolation will be broken by storm-driven dispersal and warming. *Nature Climate Change*, **8**(8), 704–708. https://doi.org/10.1038/s41558-018-0209-7

Frontier, N., **de Bettignies, F.**, **Foggo, A.**, & **Davoult, D.** (2021). Sustained productivity and respiration of degrading kelp detritus in the shallow benthos: Detached or broken, but not dead. *Marine Environmental Research*, **166**, 105277. https://doi.org/10.1016/j.marenvres.2021.105277

Frontier, N., **Mulas, M.**, **Foggo, A.**, & **Smale, D. A.** (2022). The influence of light and temperature on detritus degradation rates for kelp species with contrasting thermal affinities. *Marine Environmental Research*, **173**, 105529. https://doi.org/10.1016/j.marenvres.2021.105529

Gabry, J., **Češnovar, R.**, **Johnson, A.**, & **Bronder, S.** (2025). *cmdstanr: R Interface to 'CmdStan'*. https://mc-stan.org/cmdstanr

Gabry, J., **Simpson, D.**, **Vehtari, A.**, **Betancourt, M.**, & **Gelman, A.** (2019). Visualization in Bayesian workflow. *Journal of the Royal Statistical Society Series A: Statistics in Society*, **182**(2), 389–402. https://doi.org/10.1111/rssa.12378

Gauthier, M.-M., & **Jacobs, D. F.** (2018). Reductions in net photosynthesis and stomatal conductance vary with time since leaf detachment in three deciduous angiosperms. *Trees*, **32**(5), 1247–1252. https://doi.org/10.1007/s00468-018-1706-z

Gohel, D., **Moog, S.**, & **Heckmann, M.** (2025). *officer: Manipulation of Microsoft Word and PowerPoint documents*. https://ardata-fr.github.io/officeverse.

Gómez, I., **Wiencke, C.**, & **Thomas, D. N.** (1996). Variations in photosynthetic characteristics of the Antarctic marine brown alga *Ascoseira mirabilis* in relation to thallus age and size. *European Journal of Phycology*, **31**(2), 167–172. https://doi.org/10.1080/09670269600651341

Graiff, A., **Karsten, U.**, **Meyer, S.**, **Pfender, D.**, **Tala, F.**, & **Thiel, M.** (2013). Seasonal variation in floating persistence of detached *Durvillaea antarctica* (Chamisso) Hariot thalli. *Botanica Marina*, **56**(1), 3–14. https://doi.org/10.1515/bot-2012-0193

Graiff, A., **Pantoja, J. F.**, **Tala, F.**, & **Thiel, M.** (2016). Epibiont load causes sinking of viable kelp rafts: Seasonal variation in floating persistence of giant kelp *Macrocystis pyrifera*. *Marine Biology*, **163**(9), 191. https://doi.org/10.1007/s00227-016-2962-3

Gras, A. F., **Koch, M. S.**, & **Madden, C. J.** (2003). Phosphorus uptake kinetics of a dominant tropical seagrass *Thalassia testudinum*. *Aquatic Botany*, **76**(4), 299–315. https://doi.org/10.1016/s0304-3770(03)00069-x

Guiry, M. D., **Guiry, G. M.** (2025). *AlgaeBase*. https://www.algaebase.org

Hall, L. M., **Hanisak, M. D.**, & **Virnstein, R. W.** (2006). Fragments of the seagrasses *Halodule wrightii* and *Halophila johnsonii* as potential recruits in Indian River Lagoon, Florida. *Marine Ecology Progress Series*, **310**, 109–117. https://doi.org/10.3354/meps310109

Halm, H., **Lüder, U. H.**, & **Wiencke, C.** (2011). Induction of phlorotannins through mechanical wounding and radiation conditions in the brown macroalga *Laminaria hyperborea*. *European Journal of Phycology*, **46**(1), 16–26. https://doi.org/10.1080/09670262.2010.511277

Hamersley, M. R., **Sohm, J. A.**, **Burns, J. A.**, & **Capone, D. G.** (2015). Nitrogen fixation associated with the decomposition of the giant kelp *Macrocystis pyrifera*. *Aquatic Botany*, **125**, 57–63. https://doi.org/10.1016/j.aquabot.2015.05.003

Hanuise, D., **Lin, C.**, **Tol, S.**, **Choukroun, S.**, **Rasheed, M. A.**, **Dobbelaere, T.**, **York, P. H.**, **Smith, T. M.**, **Coles, R. G.**, **Hanert, E.**, & **Grech, A.** (2026). Experimentally derived buoyancy duration of seagrass fragments for biophysical dispersal modelling in the great barrier reef. *Marine Environmental Research*, **217**, 107937. https://doi.org/10.1016/j.marenvres.2026.107937

Harrison, P. G. (1989). Detrital processing in seagrass systems: A review of factors affecting decay rates, remineralization and detritivory. *Aquatic Botany*, **35**(3-4), 263–288. https://doi.org/10.1016/0304-3770(89)90002-8

Hemminga, M. A., **Marba, N.**, & **Stapel, J.** (1999). Leaf nutrient resorption, leaf lifespan and the retention of nutrients in seagrass systems. *Aquatic Botany*, **65**(1-4), 141–158. https://doi.org/10.1016/s0304-3770(99)00037-6

Hernández-Carmona, G., **Hughes, B.**, & **Graham, M. H.** (2006). Reproductive longevity of drifting kelp *Macrocystis pyrifera* (Phaeophyceae) in Monterey Bay, USA. *Journal of Phycology*, **42**(6), 1199–1207. https://doi.org/10.1111/j.1529-8817.2006.00290.x

Jana, S., & **Choudhuri, M. A.** (1980). Senescence in submerged aquatic angiosperms: Changes in intact and isolated leaves during aging. *New Phytologist*, **86**(2), 191–198. https://doi.org/10.1111/j.1469-8137.1980.tb03188.x

Jana, S., & **Choudhuri, M. A.** (1982). Glycolate metabolism of three submersed aquatic angiosperms during ageing. *Aquatic Botany*, **12**, 345–354. https://doi.org/10.1016/0304-3770(82)90026-2

Jiménez-Ramos, R., **Henares, C.**, **Egea, L. G.**, **Vergara, J. J.**, & **Brun, F. G.** (2023). Leaf senescence of the seagrass *Cymodocea nodosa* in Cádiz Bay, southern Spain. *Diversity*, **15**(2), 187. https://doi.org/10.3390/d15020187

Josselyn, M. N., & **Mathieson, A. C.** (1980). Seasonal influx and decomposition of autochthonous macrophyte litter in a north temperate estuary. *Hydrobiologia*, **71**(3), 197–208. https://doi.org/10.1007/bf03216236

Kar, R. K., & **Choudhuri, M. A.** (1986). Effects of light and spermine on senescence of *Hydrilla* and spinach leaves. *Plant Physiology*, **80**(4), 1030–1033. https://doi.org/10.1104/pp.80.4.1030

Kar, R. K., & **Choudhuri, M. A.** (1987). Possible mechanisms of light-induced chlorophyll degradation in senescing leaves of *Hydrilla verticillata*. *Physiologia Plantarum*, **70**(4), 729–734. https://doi.org/10.1111/j.1399-3054.1987.tb04331.x

Kar, S., **Montague, D. T.**, & **Villanueva-Morales, A.** (2021). Measurement of photosynthesis in excised leaves of ornamental trees: A novel method to estimate leaf level drought tolerance and increase experimental sample size. *Trees*, **35**(3), 889–905. https://doi.org/10.1007/s00468-021-02088-w

Kay, M. (2024). *tidybayes: tidy data and geoms for Bayesian models*. https://doi.org/10.5281/zenodo.13770114

Kennedy, J. R., & **Blain, C. O.** (2025). A systematic review of marine macroalgal degradation: Toward a better understanding of macroalgal carbon sequestration potential. *Journal of Phycology*, **61**(3), 399–432. https://doi.org/10.1111/jpy.70031

Kenworthy, W. J., **Currin, C. A.**, **Fonseca, M. S.**, & **Smith, G.** (1989). Production, decomposition, and heterotrophic utilization of the seagrass *Halophila decipiens* in a submarine canyon. *Marine Ecology Progress Series*, **51**, 277–290. https://doi.org/10.3354/meps051277

Knauer, G. A., & **Ayers, A. V.** (1977). Changes in carbon, nitrogen, adenosine triphosphate, and chlorophyll *a* in decomposing *Thalassia testudinum* leaves. *Limnology and Oceanography*, **22**(3), 408–414. https://doi.org/10.4319/lo.1977.22.3.0408

Koivikko, R., **Loponen, J.**, **Honkanen, T.**, & **Jormalainen, V.** (2005). Contents of soluble, cell-wall-bound and exuded phlorotannins in the brown alga *Fucus vesiculosus*, with implications on their ecological functions. *Journal of Chemical Ecology*, **31**(1), 195–212. https://doi.org/10.1007/s10886-005-0984-2

Küpper, F. C., **Kloareg, B.**, **Guern, J.**, & **Potin, P.** (2001). Oligoguluronates elicit an oxidative burst in the brown algal kelp *Laminaria digitata*. *Plant Physiology*, **125**(1), 278–291. https://doi.org/10.1104/pp.125.1.278

Küpper, F. C., **Müller, D. G.**, **Peters, A. F.**, **Kloareg, B.**, & **Potin, P.** (2002). Oligoalginate recognition and oxidative burst play a key role in natural and induced resistance of sporophytes of Laminariales. *Journal of Chemical Ecology*, **28**(10), 2057–2081. https://doi.org/10.1023/a:1020706129624

Küppers, U., & **Kremer, B. P.** (1978). Longitudinal profiles of carbon dioxide fixation capacities in marine macroalgae. *Plant Physiology*, **62**(1), 49–53. https://doi.org/10.1104/pp.62.1.49

Lee, D., Stan buildbot, seantalts, **Carpenter, B.**, **Morris, M.**, **Kucukelbir, A.**, **Betancourt, M.**, **Tran, D.**, **Brubaker, M.**, **Bales, B.**, et al. (2026) standev/cmdstan. *Zenodo*. https://doi.org/10.5281/zenodo.598253. Q8

Lee, K.-S., & **Dunton, K. H.** (1999). Inorganic nitrogen acquisition in the seagrass *Thalassia testudinum*: Development of a whole-plant nitrogen budget. *Limnology and Oceanography*, **44**(5), 1204–1215. https://doi.org/10.4319/lo.1999.44.5.1204

Lüder, U. H., & **Clayton, M. N.** (2004). Induction of phlorotannins in the brown macroalga *Ecklonia radiata* (Laminariales, Phaeophyta) in response to simulated herbivory – The first microscopic study. *Planta*, **218**(6), 928–937. https://doi.org/10.1007/s00425-003-1176-3

Macaya, E. C., **Boltana, S.**, **Hinojosa, I. A.**, **Macchiavello, J. E.**, **Valdivia, N. A.**, **Vasquez, N. R.**, **Buschmann, A. H.**, **Vasquez, J. A.**, **Alonso Vega, J. M.**, & **Thiel, M.** (2005). Presence of sporophylls in floating kelp rafts of macrocystis spp. (Phaeophyceae) along the Chilean Pacific coast. *Journal of Phycology*, **41**(5), 913–922. https://doi.org/10.1111/j.1529-8817.2005.00118.x

Marbà, N., **Hemminga, M. A.**, **Mateo, M. A.**, **Duarte, C. M.**, **Mass, Y. E. M.**, **Terrados, J.**, & **Gacia, E.** (2002). Carbon and nitrogen translocation between seagrass ramets. *Marine Ecology Progress Series*, **226**, 287–300. https://doi.org/10.3354/meps226287

McElreath, R. (2020) *Statistical rethinking*^{2}*: A Bayesian course with examples in R and Stan* (2nd ed. Chapman and Hall/CRC. https://doi.org/10.1201/9780429029608

Mohr, W., **Lehnen, N.**, **Ahmerkamp, S.**, **Marchant, H. K.**, **Graf, J. S.**, **Tschitschko, B.**, **Yilmaz, P.**, **Littmann, S.**, **Gruber-Vodicka, H.**, **Leisch, N.**, **Weber, M.**, **Lott, C.**, **Schubert, C. J.**, **Milucka, J.**, & **Kuypers, M. M. M.** (2021). Terrestrial-type nitrogen-fixing symbiosis between seagrass and a marine bacterium. *Nature*, **600**(7887), 105–109. https://doi.org/10.1038/s41586-021-04063-4

Moore, J. C., **Berlow, E. L.**, **Coleman, D. C.**, **Ruiter, P. C.**, **Dong, Q.**, **Hastings, A.**, **Johnson, N. C.**, **McCann, K. S.**, **Melville, K.**, **Morin, P. J.**, **Nadelhoffer, K.**, **Rosemond, A. D.**, **Post, D. M.**, **Sabo, J. L.**, **Scow, K. M.**, **Vanni, M. J.**,

& **Wall, D. H.** (2004). Detritus, trophic dynamics and biodiversity. *Ecology Letters*, **7**(7), 584–600. https://doi.org/10.1111/j.1461-0248.2004.00606.x

Müller, K. (2025). *here: a simpler way to find your files*. https://here.r-lib.org.

Neely, R. K., & **Wetzel, R. G.** (1997). Autumnal production by bacteria and autotrophs attached to *Typha latifolia* L. detritus. *Journal of Freshwater Ecology*, **12**(2), 253–267. https://doi.org/10.1080/02705060.1997.9663533

Norton, T. A. (1992). Dispersal by macroalgae. *British Phycological Journal*, **27**(3), 293–301. https://doi.org/10.1080/00071619200650271

Norton, T. A., & **Mathieson, A. C.** (1983). The biology of unattached seaweeds. In **F. E.Round** & **D. J.Chapman** (Eds.), *Progress in phycological research* (Vol. **2**, pp. 333–386).

Ochieng, C. A., & **Erftemeijer, P. L. A.** (1999). Accumulation of seagrass beach cast along the Kenyan coast: A quantitative assessment. *Aquatic Botany*, **65**(1-4), 221–238. https://doi.org/10.1016/s0304-3770(99)00042-x

Olsen, J. L., **Rouzé, P.**, **Verhelst, B.**, **Lin, Y.-C.**, **Bayer, T.**, **Collen, J.**, **Dattolo, E.**, **De Paoli, E.**, **Dittami, S.**, **Maumus, F.**, **Michel, G.**, **Kersting, A.**, **Lauritano, C.**, **Lohaus, R.**, **Töpel, M.**, **Tonon, T.**, **Vanneste, K.**, **Amirebrahimi, M.**, **Brakel, J.**, … **Van de Peer, Y.** (2016). The genome of the seagrass *Zostera marina* reveals angiosperm adaptation to the sea. *Nature*, **530**(7590), 331–335. https://doi.org/10.1038/nature16548

Olson, J. S. (1963). Energy storage and the balance of producers and decomposers in ecological systems. *Ecology*, **44**(2), 322–331. https://doi.org/10.2307/1932179

Patriquin, D., & **Knowles, R.** (1972). Nitrogen fixation in the rhizosphere of marine angiosperms. *Marine Biology*, **16**(1), 49–58. https://doi.org/10.1007/bf00347847

Pedersen, M. F., & **Borum, J.** (1992). Nitrogen dynamics of eelgrass *Zostera marina* during a late summer period of high growth and low nutrient availability. *Marine Ecology Progress Series*, **80**, 65–73. https://doi.org/10.3354/meps080065

Pedersen, M. F., **Filbee-Dexter, K.**, **Frisk, N. L.**, **Sárossy, Z.**, & **Wernberg, T.** (2021). Carbon sequestration potential increased by incomplete anaerobic decomposition of kelp detritus. *Marine Ecology Progress Series*, **660**, 53–67. https://doi.org/10.3354/meps13613

Pedersen, M. F., **Paling, E. I.**, & **Walker, D. I.** (1997). Nitrogen uptake and allocation in the seagrass *Amphibolis Antarctica*. *Aquatic Botany*, **56**(2), 105–117. https://doi.org/10.1016/s0304-3770(96)01100-x

Pedersen, T. L. (2025). patchwork: the composer of plots. https://patchwork.data-imaginist.com.

Pellikaan, G. C. (1982). Decomposition processes of eelgrass, *Zostera marina* L. *Hydrobiological Bulletin*, **16**(1), 83–92. https://doi.org/10.1007/bf02255416

Powles, J. E., **Buckley, T. N.**, **Nicotra, A. B.**, & **Farquhar, G. D.** (2006). Dynamics of stomatal water relations following leaf excision. *Plant, Cell & Environment*, **29**(5), 981–992. https://doi.org/10.1111/j.1365-3040.2005.01491.x

POWO. (2025). *Plants of the World online*. Facilitated by the Royal Botanic Gardens, Kew. https://powo.science.kew.org

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org

Rice, D. L., & **Tenore, K. R.** (1981). Dynamics of carbon and nitrogen during the decomposition of detritus derived from estuarine macrophytes. *Estuarine, Coastal and Shelf Science*, **13**(6), 681–690. https://doi.org/10.1016/s0302-3524(81)80049-7

Rodriguez, G. E., **Reed, D. C.**, & **Holbrook, S. J.** (2016). Blade life span, structural investment, and nutrient allocation in giant kelp. *Oecologia*, **182**(2), 397–404. https://doi.org/10.1007/s00442-016-3674-6

Rohatgi, A. (2022). WebPlotDigitizer. https://automeris.io

Rosch, K. L., & **Koch, M. S.** (2009). Nitrogen and phosphorus recycling by a dominant tropical seagrass (*Thalassia testudinum*) across a nutrient gradient in Florida Bay. *Bulletin of Marine Science* **84**, 1–24. https://www.ingentaconnect.com/contentone/umrsmas/bullmar/2009/00000084/00000001/art00001.

Rothäusler, E., **Gómez, I.**, **Hinojosa, I. A.**, **Karsten, U.**, **Miranda, L.**, **Tala, F.**, & **Thiel, M.** (2011). Kelp rafts in the Humboldt current: Interplay of abiotic and biotic factors limit their floating persistence and dispersal potential. *Limnology and Oceanography*, **56**(5), 1751–1763. https://doi.org/10.4319/lo.2011.56.5.1751

Rothäusler, E., **Reinwald, H.**, **López, B. A.**, **Tala, F.**, & **Thiel, M.** (2018). High acclimation potential in floating *Macrocystis pyrifera* to abiotic conditions even under grazing pressure – A field study. *Journal of Phycology*, **54**(3), 368–379. https://doi.org/10.1111/jpy.12643

Rothäusler, E., **Rugiu, L.**, **Tiihonen, T.**, & **Jormalainen, V.** (2020). It takes two to stay afloat: Interplay of morphology and physiological acclimation ensures long-term floating dispersal of the bladderwrack *Fucus vesiculosus* (Phaeophyceae, Fucales). *European Journal of Phycology*, **55**(2), 242–252. https://doi.org/10.1080/09670262.2019.1694706

Rublee, P. A., & **Roman, M. R.** (1982). Decomposition of turtlegrass (*Thalassia testudinum* Konig) in flowing sea-water tanks and litterbags: Compositional changes and comparison with natural particulate matter. *Journal of Experimental Marine Biology and Ecology*, **58**(1), 47–58. https://doi.org/10.1016/0022-0981(82)90096-x

Short, F. T., & **McRoy, C. P.** (1984). Nitrogen uptake by leaves and roots of the seagrass *Zostera marina* L. *Botanica Marina*, **27**(12), 547–556. https://doi.org/10.1515/botm.1984.27.12.547

Smith, B. D., & **Foreman, R. E.** (1984). An assessment of seaweed decomposition within a southern strait of Georgia seaweed community. *Marine Biology*, **84**(2), 197–205. https://doi.org/10.1007/bf00393005

Stafford-Bell, R. E., **Chariton, A. A.**, & **Robinson, R. W.** (2015). Prolonged buoyancy and viability of *Zostera muelleri* Irmisch ex Asch. vegetative fragments indicate a strong dispersal potential. *Journal of Experimental Marine Biology and Ecology*, **464**, 52–57. https://doi.org/10.1016/j.jembe.2014.12.014

Stapel, J., **Aarts, T. L.**, **van Duynhoven, B. H. M.**, **de Groot, J. D.**, **van den Hoogen, P. H. W.**, & **Hemminga, M. A.** (1996). Nutrient uptake by leaves and roots of the seagrass *Thalassia hemprichii* in the Spermonde Archipelago, Indonesia. *Marine Ecology Progress Series*, **134**, 195–206. https://doi.org/10.3354/meps134195

Strother, S., & **Vatta, R.** (1986). The effects of light, cycloheximide and incubation medium on the senescence of detached seagrass leaves. *Biologia Plantarum*, **28**(3), 211–218. https://doi.org/10.1007/bf02894599

Tala, F., **Gómez, I.**, **Luna-Jorquera, G.**, & **Thiel, M.** (2013). Morphological, physiological and reproductive conditions of rafting bull kelp (*Durvillaea antarctica*) in northern-central Chile (30°S). *Marine Biology*, **160**(6), 1339–1351. https://doi.org/10.1007/s00227-013-2186-8

Tala, F., **López, B. A.**, **Velásquez, M.**, **Jeldres, R.**, **Macaya, E. C.**, **Mansilla, A.**, **Ojeda, J.**, & **Thiel, M.** (2019). Long-term persistence of the floating bull kelp *Durvillaea Antarctica* from the South-East Pacific: Potential contribution to local and transoceanic connectivity. *Marine Environmental Research*, **149**, 67–79. https://doi.org/10.1016/j.marenvres.2019.05.013

Tala, F., **Velásquez, M.**, **Mansilla, A.**, **Macaya, E. C.**, & **Thiel, M.** (2016). Latitudinal and seasonal effects on short-term acclimation of floating kelp species from the South-East Pacific. *Journal of Experimental Marine Biology and Ecology*, **483**, 31–41. https://doi.org/10.1016/j.jembe.2016.06.003

Thimann, K. V., & **Satler, S. O.** (1979). Relation between leaf senescence and stomatal closure: Senescence in light. *Proceedings of the National Academy of Sciences*, **76**(5), 2295–2298. https://doi.org/10.1073/pnas.76.5.2295

Thomas, F., **Cosse, A.**, **Le Panse, S.**, **Kloareg, B.**, **Potin, P.**, & **Leblanc, C.** (2014). Kelps feature systemic defense responses: Insights into the evolution of innate immunity in multicellular eukaryotes. *New Phytologist*, **204**(3), 567–576. https://doi.org/10.1111/nph.12925

Trevathan-Tackett, S. M., **Jeffries, T. C.**, **Macreadie, P. I.**, **Manojlovic, B.**, & **Ralph, P.** (2020). Long-term decomposition captures key steps in microbial breakdown of seagrass litter. *Science of the Total Environment*, **705**, 135806. https://doi.org/10.1016/j.scitotenv.2019.135806

Twigg, I. M., **Baltar, F.**, **Hall, J. R.**, & **Hepburn, C. D.** (2020). Revealing hydrogen peroxide as an external stressor in macrophyte-dominated coastal ecosystems. *Oecologia*, **193**(3), 583–591. https://doi.org/10.1007/s00442-020-04690-0

Vähätalo, A., **Søndergaard, M.**, **Schlüter, L.**, & **Markager, S.** (1998). Impact of solar radiation on the decomposition of detrital leaves of eelgrass *Zostera marina*. *Marine Ecology Progress Series*, **170**, 107–117. https://doi.org/10.3354/meps170107

van den Brand, T. (2025). *ggh4x: hacks for 'ggplot2'*. https://github.com/teunbrand/ggh4x.

Vandendriessche, S., **Vincx, M.**, & **Degraer, S.** (2007). Floating seaweed and the influences of temperature, grazing and clump size on raft longevity – A microcosm study. *Journal of Experimental Marine Biology and Ecology*, **343**(1), 64–73. https://doi.org/10.1016/j.jembe.2006.11.010

Viana, I. G., **Saavedra-Hortúa, D. A.**, **Mtolera, M.**, & **Teichberg, M.** (2019). Different strategies of nitrogen acquisition in two tropical seagrasses under nitrogen enrichment. *New Phytologist*, **223**(3), 1217–1229. https://doi.org/10.1111/nph.15885

Waycott, M., **Biffin, E.**, & **Les, D. H.** (2018). Systematics and evolution of Australian seagrasses in a global context. In **A. W. D. Larkum**, **G. A. Kendrick**, & **P. J. Ralph** (Eds.), *Seagrasses of Australia: Structure, ecology and conservation* (pp. 129–154). Springer International Publishing. https://doi.org/10.1007/978-3-319-71354-0_5

Weatherall, E. J., **Ostrowski, A.**, **Lekammudiyanse, M. U.**, **Hendry, R. A.**, & **Jackson, E. L.** (2025). Re-attachment and viability of vegetative propagules of subtropical intertidal seagrasses. *Marine Ecology Progress Series*, **767**, 27–40. https://doi.org/10.3354/meps14917

Weinberger, F., & **Friedlander, M.** (2000). Response of *Gracilaria conferta* (Rhodophyta) to oligoagars results in defense against agar-degrading epiphytes. *Journal of Phycology*, **36**(6), 1079–1086. https://doi.org/10.1046/j.1529-8817.2000.00003.x

Weinberger, F., **Friedlander, M.**, & **Hoppe, H.** (1999). Oligoagars elicit a physiological response in *Gracilaria conferta* (Rhodophyta). *Journal of Phycology*, **35**(4), 747–755. https://doi.org/10.1046/j.1529-8817.1999.3540747.x

White, L. J., & **Norkko, A.** (2025). A path towards appropriate degradation experiments for assessing carbon sequestration potential of macroalgae. *Journal of Ecology*, **113**(10), 2730–2743. https://doi.org/10.1111/1365-2745.70107

Wickham, H., **Averick, M.**, **Bryan, J.**, **Chang, W.**, **McGowan, L. D.**, **François, R.**, **Grolemund, G.**, **Hayes, A.**, **Henry, L.**, **Hester, J.**, **Kuhn, M.**, **Pedersen, T.**, **Miller, E.**, **Bache, S.**, **Müller, K.**, **Ooms, J.**, **Robinson, D.**, **Seidel, D.**, **Spinu, V.**, … **Yutani, H.** (2019). Welcome to the Tidyverse. *Journal of Open Source Software*, **4**(43), 1686. https://doi.org/10.21105/joss.01686

Wilke, C. O. (2021). *ggridges: Ridgeline plots in ggplot2*. https://doi.org/10.32614/CRAN.package.ggridges

Wolodzko, T. (2023). *extraDistr: Additional univariate and multivariate distributions*. https://doi.org/10.32614/CRAN.package.extraDistr

Wright, L. S. (2025). *Detrital photosynthesis (Dataset)*. https://doi.org/10.5281/zenodo.15054888

Wright, L. S. (2026a). *Macroalgal exponential decay (Dataset)*. https://doi.org/10.5281/zenodo.18169931.

Wright, L. S. (2026b). A model of macroalgal decomposition. *Ecological Modelling*, **519**, 111549. https://doi.org/10.1016/j.ecolmodel.2026.111549

Wright, L. S., & **Foggo, A.** (2021). Photosynthetic pigments of co-occurring Northeast Atlantic *Laminaria* spp. are unaffected by decomposition. *Marine Ecology Progress Series*, **678**, 227–232. https://doi.org/10.3354/meps13886

Wright, L. S., & **Kregting, L.** (2023). Genus-specific response of kelp photosynthetic pigments to decomposition. *Marine Biology*, **170**(11), 144. https://doi.org/10.1007/s00227-023-04289-y

Wright, L. S., **Pessarrodona, A.**, & **Foggo, A.** (2022). Climate-driven shifts in kelp forest composition reduce carbon sequestration potential. *Global Change Biology*, **28**(18), 5514–5531. https://doi.org/10.1111/gcb.16299

Wright, L. S., **Simpkins, T.**, **Filbee-Dexter, K.**, & **Wernberg, T.** (2024). Temperature sensitivity of detrital photosynthesis. *Annals of Botany*, **133**(1), 17–28. https://doi.org/10.1093/aob/mcad167